

Lakhan Jadhwant

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PROFILE

AI Engineer with hands-on experience building AI/ML and Generative AI applications using Python, LLMs, RAG, LangChain, and FastAPI. Experienced in developing AI-powered solutions for document processing, automation, and real-world applications. Strong foundation in machine learning, software development, and AI automation.

WORK EXPERIENCE

AI Engineer Intern, Novametrics AI 01/2026 – 07/2026

- Developed a drone-based computer vision segmentation pipeline using **SAM 3** to detect 6+ object classes from aerial video footage
- Developed an RF-DETR-based road defect detection system to identify potholes, cracks, patches, and scaling from road imagery, automating defect identification for infrastructure inspection.
- Experimented with **3D Gaussian Splatting** to reconstruct 3D scenes from Insta360 video footage of interiors, roads, buildings, and buses.

Data Scientist Intern, Cornerstone Solutions 11/2025 – 12/2025

- Optimized an ANPR pipeline by integrating **RF-DETR** for license-plate detection with **PaddleOCR** for text recognition, improving automated plate-reading reliability.
- Worked on developing a windmill inspection system using **computer vision** for automated fault detection and structural assessment.

AI/ML intern, Novel Group 06/2025 – 08/2025

- Engineered and trained a **RASA chatbot** that resolved a majority of healthcare queries, cutting manual workload and improving response speed by ~80%.
- Built an **AI-powered resume evaluation tool** (Streamlit, FastAPI, LangChain, MongoDB) that analyzed resumes, delivered compatibility scores, and provided AI-driven feedback.
- Designed and executed an **outbound sales pipeline**, boosting qualified lead generation by ~70%.

PROJECTS

Multi Document RAG Chatbot 📄

- Built a full-stack multi-document RAG chatbot using **FastAPI** and **Streamlit**, enabling users to query uploaded documents through context-aware AI responses.
- Implemented a **LangChain** retrieval pipeline with Gemini embeddings and **Pinecone** vector search to efficiently retrieve relevant document context for LLM-generated responses.
- Integrated **Groq** LLM API to deliver low-latency, context-aware responses for real-time document question answering.

Face Recognition Attendance System 📄

- Built a **Streamlit** web application that automates classroom attendance using **Deep Learning** and **GCP**.
- Implemented **DeepFace** with **RetinaFace** for face detection and **FaceNet512 embeddings** for face recognition.
- Used **scikit-learn** to compute **cosine similarity** between face embeddings for accurate matching.
- Integrated **gspreed** to update **Google Sheets** in real-time, enabling instant attendance reports.
- Eliminated manual entry and reduced processing time by ~90%, making the system scalable for large classes.

SKILLS

- Programming:** Python, C++
- Databases:** MySQL, MongoDB, Vector Databases (Pinecone, ChromaDB, FAISS)
- Libraries:** NumPy, Pandas, Matplotlib, Seaborn, NLTK, spaCy, Scikit-learn, PyTorch, TensorFlow
- AI/ML & GenAI:** Machine Learning, Deep Learning, NLP, Computer Vision, RAG, LLMs, Prompt Engineering
- Frameworks & Platforms:** LangChain, LangGraph, FastAPI, Streamlit, Rasa, Dialogflow, Roboflow, AI Agents
- Cloud & DevOps:** Google Cloud, Docker
- Version Control:** Git, GitHub, Bitbucket

EDUCATION

B.Tech + M.Tech (Dual Degree) in AI and Data Science, Devi Ahilya Vishwavidyalaya 2021 – 2026
8.76 CGPA

Higher Secondary, School for Excellence 2020 – 2021
93.8%

CERTIFICATES

- PW - Data Science Master's 📄
- Coursera - Data Analytics 📄
- Oracle Cloud Infrastructure 📄